

COMPUTER SCIENCE TRIPOS Part IB – 2023 – Paper 7

8 Further Graphics (aco41)

Rendering a plane (a) We are given a scene with a point light source at a point $\mathbf{p}$ that emits light of radiance L in all directions, and a plane with equation $\mathbf{n}^T(\mathbf{x} - \mathbf{c}) = 0$ and BRDF f_r . We send a ray from the camera sensor with origin $\mathbf{o}$ and direction $\mathbf{d}$ such that each point on the ray is $\mathbf{r}(t) = \mathbf{o} + t\mathbf{d}$ for $t > 0$.

- (i) What is the intersection point of the ray and the plane? [2 marks]
- (ii) What is the incoming light direction? [1 mark]
- (iii) What is the direction of the outgoing reflected light towards the sensor origin $\mathbf{o}$? [1 mark]
- (iv) What is the radiance of the light reflected towards the sensor origin $\mathbf{o}$? [2 marks]

Answer:

- (i) The intersection of the ray with the plane is given by $\mathbf{n}^T(\mathbf{o} + t\mathbf{d} - \mathbf{c}) = 0$. [1 mark]
Solving for t , we get $t^* = \mathbf{n}^T(\mathbf{c} - \mathbf{o})/\mathbf{n}^T\mathbf{d}$ and the intersection point is $\mathbf{x} = \mathbf{o} + t^*\mathbf{d}$, assuming $\mathbf{n}^T\mathbf{d}$ is not zero. [1 mark]
 - (ii) As we have a single point light source, the incoming light direction is $\vec{\omega}_i = (\mathbf{p} - \mathbf{x})/||\mathbf{p} - \mathbf{x}||$ and no light is received in other directions. [1 mark]
 - (iii) The outgoing light direction is $\vec{\omega}_o = -\mathbf{d}/||\mathbf{d}||$. [1 mark]
 - (iv) The outgoing light simplifies as $L_o(\mathbf{x}, \vec{\omega}_o) = \int_{H^2} f_r(\mathbf{x}, \vec{\omega}_i, \vec{\omega}_o) L_i(\mathbf{x}, \vec{\omega}_i) \cos \theta_i d\vec{\omega}_i = f_r(\mathbf{x}, \vec{\omega}_i, \vec{\omega}_o) L \cos \theta_i$. [1 mark] Finally, $\cos \theta_i = \mathbf{n}^T(\mathbf{p} - \mathbf{x})/(||\mathbf{n}|| ||\mathbf{p} - \mathbf{x}||)$ [1 mark], assuming positive dot product.
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LBS for an implicit surface

(b) A surface represented by an implicit function $f(\mathbf{x})$ is transformed by linear blend skinning such that each point $\mathbf{x}$ is mapped to $\mathbf{x}'$. There are two bones with transformation matrices $\mathbf{T}_1$ and $\mathbf{T}_2$ and corresponding blending weights $w_1(\mathbf{x})$ and $w_2(\mathbf{x}) = 1 - w_1(\mathbf{x})$. In addition, we are given $s_1(\mathbf{x}') = w_1(\mathbf{x})$ and $s_2(\mathbf{x}') = w_2(\mathbf{x})$.

- (i) Write the expression for $\mathbf{x}'$. [1 mark]
- (ii) Simplify the expression for $\mathbf{x}'$ assuming no rotations are stored on the bones. Modify the expression further for the case where the weights are not summing up to 1. [2 marks]
- (iii) What is the implicit function of the transformed surface assuming all blended rotations are invertible? Simplify as much as possible. [3 marks]
- (iv) What is the implicit function of the transformed surface if there are no rotations stored on the bones? Simplify as much as possible. [2 marks]

Answer:

- (i) A point on the new surface is given by: $\mathbf{x}' = \mathbf{T}(\mathbf{x})\mathbf{x}$, where $\mathbf{T}(\mathbf{x}) = \mathbf{T}_1 w_1(\mathbf{x}) + \mathbf{T}_2 w_2(\mathbf{x})$.
- (ii) In this case, the expression simplifies to $\mathbf{x}' = \mathbf{x} + \mathbf{t}_1 w_1(\mathbf{x}) + \mathbf{t}_2 w_2(\mathbf{x})$ for translations $\mathbf{t}_1$ and $\mathbf{t}_2$. [1 mark] This is only true when $w_1(\mathbf{x}) + w_2(\mathbf{x}) = 1$, otherwise we get $\mathbf{x}' = \mathbf{x} + (\mathbf{t}_1 w_1(\mathbf{x}) + \mathbf{t}_2 w_2(\mathbf{x})) / (w_1(\mathbf{x}) + w_2(\mathbf{x}))$. [1 mark]

- (iii) To make the transformation $\mathbf{T}(\mathbf{x})$ a function of $\mathbf{x}'$, we can rewrite it as $\mathbf{T}(\mathbf{x}) = \mathbf{T}(\mathbf{x}') = \mathbf{T}_1 s_1(\mathbf{x}') + \mathbf{T}_2 s_2(\mathbf{x}')$. [1 mark] We can then write $\mathbf{x} = \mathbf{T}^{-1}(\mathbf{x}')\mathbf{x}'$, assuming $\mathbf{T}$ is invertible, which is always true as the blended rotations are invertible. [1 mark] We know that $f(\mathbf{x}) = 0$ for a point on the surface. Substituting the expression for $\mathbf{x}$, we get $f(\mathbf{T}(\mathbf{x}')^{-1}\mathbf{x}') = 0$. [1 mark]
- (iv) In this case, $\mathbf{T}^{-1}(\mathbf{x}') = -\mathbf{t}_1 s_1(\mathbf{x}') - \mathbf{t}_2 s_2(\mathbf{x}')$ for translations $\mathbf{t}_1$ and $\mathbf{t}_2$ [1 mark] and thus $f(\mathbf{x}' - \mathbf{t}_1 s_1(\mathbf{x}') - \mathbf{t}_2 s_2(\mathbf{x}')) = 0$. [1 mark]
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Quaternions for
spatial rotation
representation

- (c) In this question, we will think about how quaternions represent spatial rotations.
- (i) Write the form of a general quaternion and the form of a quaternion representing rotations. [2 marks]
- (ii) Prove that a quaternion $\mathbf{q}$ and $-\mathbf{q}$ represents the same rotation. [2 marks]
- (iii) Given a quaternion $\mathbf{q}$ with no real part, what is $\mathbf{q}\mathbf{q}$? [2 marks]
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Answer:

- (i) General form: $\mathbf{q} = a + bi + cj + dk$ [1 mark], for spatial rotations: $\mathbf{q} = \cos(\theta/2) + \sin(\theta/2)\mathbf{s}$ for rotation of angle θ around the axis $\mathbf{s}$. [1 mark]
- (ii) Rotation of a vector $\mathbf{v}$ by a quaternion $\mathbf{q}$ is given by $\mathbf{v}' = \mathbf{q}\mathbf{v}\mathbf{q}^*$. [1 mark] Rotation by $-\mathbf{q}$ is thus $\mathbf{v}'' = -\mathbf{q}\mathbf{v}(-\mathbf{q})^* = \mathbf{q}\mathbf{v}\mathbf{q}^* = \mathbf{v}'$ since $(-\mathbf{q})^* = -\mathbf{q}^*$. [1 mark]
- (iii) A quaternion with no real part means $\mathbf{q} = bi + cj + dk$. Utilising the multiplication properties of i, j, k [1 mark], we get $\mathbf{q}\mathbf{q} = -(b^2 + c^2 + d^2) = -\|\mathbf{q}\|^2$ [1 mark].
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